Group II Row IV: DGLAP, running coupling and finite terms, step by step

Your equation (Group II Row IV, no c.c.). Conventions:

- $a \equiv \frac{k^+}{\vee}$, $\alpha_s = \frac{g^2}{4\pi}$, $\beta_0 = \frac{11}{3}N_c$ ($n_f = 0$), $c_0 = 2e^{-\gamma_E}$, $\Delta \equiv y' - y$.
- $\ell_\Delta(\mu) \equiv 2\gamma_E + \log\frac{(y'-y)^2\mu^2}{4}$.
- $d_\perp = 2 - \epsilon$ **everywhere; no step sets** $d_\perp = 2$**.**

Blue = new, red = removed (with respect to the previous version of this note).

1 Answer

- **No** $\log\frac{\vee}{k^+}$**.** You are right: none is left. The previous version showed $-\log\frac{\vee}{k^+}H_4$ because I had done the y' integral of your subtraction line (L2) on its own.
  - Your line (L1) contains $+\log\frac{\vee}{k^+}$ inside its ξ integral, from $\xi \to \frac{k^+}{\vee}$.
  - Your line (L2) is exactly the subtraction that removes it.
  - Separately the two are $\pm\log\frac{\vee}{k^+}$; together they are finite for $\vee \to \infty$.

  Here (L2) is kept as you wrote it, inside the DGLAP term, and no $\log\frac{\vee}{k^+}$ appears (Step 7). The $+\log\frac{\vee}{k^+}H_4$ of your BFKL list is the counterpart of this subtraction, so nothing is counted twice.
- **DGLAP.** Your bracket is *exactly* $-\frac{2}{\epsilon} - \frac{d_\perp}{2}\ell_\Delta(\mu)$: the $+1$ of $-\frac{d_\perp}{\epsilon}$ cancels your -1. The pole of the row is $\frac{\alpha_s}{2\pi}(-\frac{2}{\epsilon})P^{(r)}_{gg,+} \otimes \mathrm{LO}$, with $P^{(r)}_{gg,+} = P_{gg} - \frac{\beta_0}{2}\delta(1-\xi)$.
  - The P_{gg} part goes into $D_{h/g}(\zeta, \mu_F)$, which obeys $\mu_F^2\partial_{\mu_F^2}D = \frac{\alpha_s}{2\pi}P_{gg} \otimes D$.
- **Running coupling.** The $-\frac{\beta_0}{2}\delta(1-\xi)$ part, together with the $\overline{MS}$ counterterm of the coupling of LO, leaves

$$\frac{\alpha_s\beta_0}{4\pi}\,\mathrm{LO}\Big[\frac{d_\perp}{2}\,\ell_\Delta(\mu_R)\Big] = \frac{\alpha_s\beta_0}{4\pi}\,\mathrm{LO}\Big[\frac{d_\perp}{2}\log\frac{(y'-y)^2\mu_R^2}{c_0^2}\Big].$$

  This is the running of the coupling of LO with the transverse distance $|y'-y|$, conjugate to $\mathbf{k}$. At $\epsilon \to 0$ it turns $\alpha_s(\mu_R)$ into $\alpha_s(c_0/|y'-y|)$.
- **Finite terms.** Eq. (4.1) gives all of them in your notation:
  - the DGLAP term with your subtraction (L2);
  - your (L3);
  - the running-coupling term;
  - your (L5).

  In practice: your row with $-\frac{d_\perp}{\epsilon}$ and -1 removed (they add up to exactly $-\frac{2}{\epsilon}$, which goes into $D_{h/g}$ and the coupling) and $\mu_{\overline{MS}} \to \mu_F$, plus the two $\frac{11}{12}$ terms.

Other β_0 terms. Group I Row II part 1 ($\frac{11}{3}$) and Group I Row V $d^3N_{2b\,\mathrm{UV}}$ ($-\frac{11}{6}$) have β_0 terms of their own. These cancel each other exactly. Keep both or drop both: in your latest `Other_terms.tex`, Row II part 1 is commented out but Row V $2b$ UV is not.

2 Notation

$$\mathrm{LO}[f] \equiv \frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\, f\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\Big[U^{e'c}(y') - U^{e'c}(x')\Big]\,[U^{ec}(y) - U^{ce}(x)]\,\rho^{e'}(x')\rho^e(x), \tag{2.1}$$

$\mathrm{LO} \equiv \mathrm{LO}[1]$. $\mathrm{LO}_\xi[f]$: the same with $e^{-ik\cdot(y'-y)} \to e^{-ik\cdot(y'-y)\frac{1}{\xi}}$, so $\mathrm{LO}_\xi = \frac{1}{\xi}\frac{d^3N_{\mathrm{LO}}}{d^3k}(\frac{k^+}{\xi}, \frac{\mathbf{k}}{\xi})$ ($d^3k = dk^+d^2\mathbf{k}$). $\mathrm{LO}^{(y)}_\xi[f]$: the same as $\mathrm{LO}_\xi[f]$ with $\frac{(x'-y')}{(x'-y')^2} \to \frac{(x'-y)}{(x'-y)^2}$ and $U^{e'c}(y') \to U^{e'c}(y)$, i.e. the $y' = y$ value used in your line (L2).

$$P^{(r)}_{gg,+}(\xi) = 2N_c\Big[\frac{\xi}{(1-\xi)_+} + \frac{1-\xi}{\xi} + \xi(1-\xi)\Big], \qquad P_{gg}(\xi) = P^{(r)}_{gg,+}(\xi) + \frac{\beta_0}{2}\,\delta(1-\xi), \tag{2.2}$$

with the plus distribution of $[0,1]$: $\int_a^1 d\xi \frac{g(\xi)}{(1-\xi)_+} = \int_a^1 d\xi \frac{g(\xi)-g(1)}{1-\xi} + g(1)\log(1-a)$.

3 Step by step

Step 0: your row, line by line.

$$
\begin{aligned}
(L1)\quad & \frac{g^2}{4\pi^2}\,\frac{N_c}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{1}{k^+}\int_{x,y,x',y'}\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[-\frac{d_\perp}{\epsilon}-d_\perp\,\gamma_E-\frac{d_\perp}{2}\,\log\left(\frac{(y'-y)^2\,\mu_{\overline{MS}}^2}{4}\right)-1\right]\\
& \left[\int_{\frac{k^+}{\vee}}^1\frac{d\xi}{\xi^2}\,e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\left[\xi\,(1-\xi)+\frac{(1-\xi)}{\xi}+\xi+\frac{\xi^2}{(1-\xi)_+}\right]\right]\times\Big[U^{e'c}(y')-U^{e'c}(x')\Big]\times\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^{e}(x)\\
(L2)\quad & -\frac{g^2}{4\pi^2}\,\frac{N_c}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{1}{k^+}\int_{x,y,x',y'}\frac{(x-y)\cdot(x'-y)}{(x-y)^2\,(x'-y)^2}\left[-\frac{d_\perp}{2}\,\log\left(\frac{(y'-y)^2\,\mu_{\overline{MS}}^2}{4}\right)\right]\\
& \left[\int_{\frac{k^+}{\vee}}^1\frac{d\xi}{\xi^2}\,e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\left[\frac{(1-\xi)}{\xi}\right]\right]\times\Big[U^{e'c}(y)-U^{e'c}(x')\Big]\times\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^{e}(x)\\
(L3)\quad & -\frac{1}{(2\pi)^3}\,\frac{g^4\,N_c}{8\pi^5}\,\frac{1}{k^+}\left(1-\frac{k^+}{\vee}\right)\int_{x,x',z,w,w'}e^{-ik\cdot(w'-w)}\,\frac{(w-z)\cdot(w'-z)}{(w-z)^2\,(w'-z)^2}\,\frac{(x-z)\cdot(x'-z)}{(x-z)^2\,(x'-z)^2}\\
& \Big[U^{e'c}(z)-U^{e'c}(x')\Big]\times\left[U^{ec}(z)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^{e}(x)\\
(L4)\quad & +\frac{g^2}{4\pi^2}\,\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{N_c}{k^+}\log\left(1-\frac{k^+}{\vee}\right)\int_{x,y,x',y'}e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[-\frac{d_\perp}{\epsilon}-d_\perp\,\gamma_E\right.\\
& \left.-\frac{d_\perp}{2}\,\log\left(\frac{(y'-y)^2\,\mu_{\overline{MS}}^2}{4}\right)-1\right]\Big[U^{e'c}(y')-U^{e'c}(x')\Big]\times\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^{e}(x)\\
(L5)\quad & +\frac{g^2}{4\pi^2}\,\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{N_c}{k^+}\int_{\frac{k^+}{\vee}}^1 d\xi\,\frac{(1-\xi)}{\xi}\int_{x,y,x',y'}e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\left[\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}-\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2\,(x'-y')^2\,(y'-y)^2}\,d_\perp\right]\times\\
& \Big[U^{e'c}(y')-U^{e'c}(x')\Big]\times\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^{e}(x)
\end{aligned}
\tag{3.1}
$$

Step 1: units of LO. $\frac{g^2}{4\pi^2}=\frac{\alpha_s}{\pi}$. The x,y,x',y' integrands of (L1), (L4) are those of (2.1), (L1) with $e^{-ik\cdot\Delta}\to e^{-ik\cdot\Delta/\xi}$. With $\mathcal{B}_\Delta$ your bracket:

$$
(\mathrm{L1})=\frac{\alpha_s N_c}{\pi}\int_a^1\frac{d\xi}{\xi^2}\Big[\xi(1-\xi)+\frac{1-\xi}{\xi}+\xi+\frac{\xi^2}{(1-\xi)_+}\Big]_{\text{yours}}\mathrm{LO}_\xi[\mathcal{B}_\Delta],\qquad (\mathrm{L4})=\frac{\alpha_s N_c}{\pi}\log(1-a)\,\mathrm{LO}[\mathcal{B}_\Delta].
\tag{3.2}
$$

Step 2: the bracket, exactly. With $d_\perp=2-\epsilon$, $-\frac{d_\perp}{\epsilon}=-\frac{2}{\epsilon}+1$. So, with no expansion in ϵ:

$$
\mathcal{B}_\Delta=-\frac{d_\perp}{\epsilon}-d_\perp\gamma_E-\frac{d_\perp}{2}\log\frac{\Delta^2\mu^2}{4}-1=-\frac{2}{\epsilon}-\frac{d_\perp}{2}\Big[2\gamma_E+\log\frac{\Delta^2\mu^2}{4}\Big]=-\frac{2}{\epsilon}-\frac{d_\perp}{2}\,\ell_\Delta(\mu).
\tag{3.3}
$$

The $+1$ of $-\frac{d_\perp}{\epsilon}$ and your -1 cancel exactly (`make_note.py`, sympy).

Step 3: (L4) completes the plus distribution. Your prescription, $\int\frac{d\xi}{\xi^2}\frac{\xi^2}{(1-\xi)_+}G=\int\frac{d\xi}{1-\xi}[G(\xi)-G(1)]$, has no $\log(1-a)$. On $[a,1]$:

$$
\int_a^1\frac{d\xi}{\xi^2}\Big[\xi+\frac{\xi^2}{(1-\xi)_+}\Big]_{\text{yours}}G=\int_a^1\frac{d\xi}{\xi^2}\frac{\xi}{(1-\xi)_+}G-\log(1-a)\,G(1),
$$

with $\mathrm{LO}_{\xi=1}=\mathrm{LO}$ (`other_check/checks.py`, item 3). (L4) adds the missing term:

$$
(\mathrm{L1})+(\mathrm{L4})=\frac{\alpha_s}{2\pi}\int_a^1\frac{d\xi}{\xi^2}\,P^{(r)}_{gg,+}(\xi)\,\mathrm{LO}_\xi\Big[-\frac{2}{\epsilon}-\frac{d_\perp}{2}\ell_\Delta(\mu)\Big].
\tag{3.4}
$$

Step 4: split $P^{(r)}_{gg,+}$. With $P^{(r)}_{gg,+}=P_{gg}-\frac{\beta_0}{2}\delta(1-\xi)$:

$$
(\mathrm{L1})+(\mathrm{L4})=\underbrace{\frac{\alpha_s}{2\pi}\int_a^1\frac{d\xi}{\xi^2}\,P_{gg}(\xi)\,\mathrm{LO}_\xi\Big[-\frac{2}{\epsilon}-\frac{d_\perp}{2}\ell_\Delta(\mu)\Big]}_{\text{DGLAP}}+\underbrace{\frac{\alpha_s\beta_0}{4\pi}\,\mathrm{LO}\Big[\frac{2}{\epsilon}+\frac{d_\perp}{2}\ell_\Delta(\mu)\Big]}_{\text{running coupling}}.
\tag{3.5}
$$

Step 5: DGLAP. $\int_a^1\frac{d\xi}{\xi^2}P\,\mathrm{LO}_\xi=\int_{k^+/\vee}^1\frac{d\xi}{\xi^3}P(\xi)\frac{d^3N_{\mathrm{LO}}}{d^3k}(\frac{k^+}{\xi},\frac{\mathbf{k}}{\xi})$. This is a gluon of momentum $\frac{k}{\xi}$ that turns into one with the fraction ξ. For a hadron, $\frac{d^3N_h}{d^3k}=\int_{k^+/\vee}^1\frac{d\zeta}{\zeta^3}D^{\mathrm{bare}}_{h/g}(\zeta)\frac{d^3N_g}{d^3k}(\frac{k^+}{\zeta},\frac{\mathbf{k}}{\zeta})$.

- With $z=\zeta\xi$ the pole becomes $\frac{\alpha_s}{2\pi}(-\frac{2}{\epsilon})\int_a^1\frac{dz}{z^3}\frac{d^3N_{\mathrm{LO}}}{d^3k}(\frac{k}{z})[P_{gg}\otimes D](z)$, with $[P\otimes D](z)=\int_z^1\frac{d\zeta}{\zeta}P(\frac{z}{\zeta})D(\zeta)$. This needs the plus distribution of Step 3 (`other_check/checks.py`, item 5).

- The $\overline{MS}$ counterterm of D at μ_F is

$$D_{h/g}^{\rm bare} = D_{h/g}(\mu_F) + \frac{\alpha_s}{2\pi}\Big[\frac{2}{\epsilon} + \frac{d_\perp}{2}\log\frac{\mu^2}{\mu_F^2}\Big] P_{gg}\otimes D_{h/g}(\mu_F), \qquad \mu_F^2\frac{\partial D_{h/g}}{\partial \mu_F^2} = \frac{\alpha_s}{2\pi}P_{gg}\otimes D_{h/g}. \tag{3.6}$$

  For $\mu_F = \mu$ it is the pure pole. The log of μ/μ_F carries the same $\frac{d_\perp}{2}$ as your bracket.

With $\ell_\Delta(\mu) = \ell_\Delta(\mu_F) + \log\frac{\mu^2}{\mu_F^2}$, the DGLAP term of (3.5) plus (3.6) leaves, exactly,

$$-\frac{\alpha_s}{2\pi}\int_{k^+/\vee}^1 \frac{d\xi}{\xi^2}\, P_{gg}(\xi)\,\mathrm{LO}_\xi\Big[\frac{d_\perp}{2}\ell_\Delta(\mu_F)\Big]. \tag{3.7}$$

Step 6: running coupling. The $\overline{MS}$ counterterm of the coupling of LO at μ_R is $-\frac{\alpha_s\beta_0}{4\pi}\big[\frac{2}{\epsilon} + \frac{d_\perp}{2}\log\frac{\mu^2}{\mu_R^2}\big]\mathrm{LO}$. Added to the second term of (3.5):

$$\boxed{\frac{\alpha_s\beta_0}{4\pi}\,\mathrm{LO}\Big[\frac{d_\perp}{2}\log\frac{(y'-y)^2\mu_R^2}{c_0^2}\Big].} \tag{3.8}$$

The pole and μ cancel exactly. Since $\alpha_s(Q) = \alpha_s(\mu_R)[1 - \frac{\alpha_s\beta_0}{4\pi}\log\frac{Q^2}{\mu_R^2}]$, at $\epsilon\to 0$ this is $\mathrm{LO}|_{\alpha_s(\mu_R)} \to \mathrm{LO}[\alpha_s(c_0/|y'-y|)]$: the coupling of LO runs with $|y'-y|$, and μ_R drops out.

- Split at μ_F:

$$(3.8) = \frac{\alpha_s\beta_0}{4\pi}\frac{d_\perp}{2}\log\frac{\mu_R^2}{\mu_F^2}\mathrm{LO} + \frac{\alpha_s\beta_0}{4\pi}\mathrm{LO}\Big[\frac{d_\perp}{2}\ell_\Delta(\mu_F)\Big].$$

  The first term is the renormalization-group term of `Other_terms_analysis.pdf`.

- The bookkeeping of Steps 5–6 is checked symbolically with $d_\perp = 2-\epsilon$ kept (`make_note.py`).

Step 7: (L2) and the absence of $\log\frac{\vee}{k^+}$. Only the $\frac{1-\xi}{\xi}$ term of P_{gg} matters at small ξ.

- **Where the hidden log is.** In (3.7) this term gives $\int_a \frac{d\xi}{\xi^2}\frac{1-\xi}{\xi}\mathrm{LO}_\xi[\dots]$. For $\xi\to 0$ the phase $e^{-ik\cdot\Delta/\xi}$ restricts $|y'-y| \lesssim \xi/|\mathbf{k}|$, so $\mathrm{LO}_\xi \propto \xi^2$. The integral then behaves as $\int_a \frac{d\xi}{\xi} = \log\frac{\vee}{k^+}$.

- **How (L2) removes it.** (L2) is the same term at $y' = y$, i.e. $\mathrm{LO}_\xi^{(y)}$.

  – Its μ and γ_E parts multiply $\int_{y'} e^{-ik\cdot(y'-y)/\xi} \propto \delta^{(2)}(\mathbf{k})$ and vanish for $\mathbf{k}\neq 0$. So exactly

$$(\mathrm{L2}) = +\frac{\alpha_s}{2\pi}\,2N_c\int_a^1 \frac{d\xi}{\xi^2}\frac{1-\xi}{\xi}\,\mathrm{LO}_\xi^{(y)}\Big[\frac{d_\perp}{2}\ell_\Delta(\mu_F)\Big]. \tag{3.9}$$

  – Combined with (3.7), the small-ξ integrand becomes $\frac{1-\xi}{\xi}\big(\mathrm{LO}_\xi - \mathrm{LO}_\xi^{(y)}\big)$.

  – This difference contains $U(y') - U(y)$ and $\frac{x'-y'}{(x'-y')^2} - \frac{x'-y}{(x'-y)^2}$, both $O(y'-y) = O(\xi)$. The ξ integral therefore converges at $\xi\to 0$, and the sum has a finite limit for $\vee\to\infty$.

The DGLAP term of the row is therefore

$$\boxed{\left.\frac{d^3N}{d^3k}\right|_{\rm DGLAP} = -\frac{\alpha_s}{2\pi}\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}\Big\{P_{gg}(\xi)\,\mathrm{LO}_\xi\Big[\frac{d_\perp}{2}\ell_\Delta(\mu_F)\Big] - 2N_c\frac{1-\xi}{\xi}\,\mathrm{LO}_\xi^{(y)}\Big[\frac{d_\perp}{2}\ell_\Delta(\mu_F)\Big]\Big\}} \tag{3.10}$$

with no $\log\frac{\vee}{k^+}$. (Doing the y' integral of (L2) on its own, with the two-dimensional Fourier transform, gives (L2) $= -\frac{d_\perp}{2}\big[\log\frac{\vee}{k^+} - (1-\frac{k^+}{\vee})\big]H_4$, as in the previous version. That is the same subtraction written out: its $\log\frac{\vee}{k^+}$ cancels the hidden one of the first term, so it must not be listed separately.)

Step 8: (L3) and (L5). Both have no pole and are kept as you wrote them, with $d_\perp = 2-\epsilon$.

- (L3) $= -(1-\frac{k^+}{\vee})H_4$ is bounded for every $\vee$.

- (L5): the Δ angular average of its bracket vanishes (not DGLAP), and its measure is $d\xi$, so it is finite for $\vee\to\infty$.

4 Result in your notation

After $D_{h/g}(\zeta,\mu_F)$ takes the P_{gg} pole and the coupling counterterm takes the β_0 pole, your row is, exactly ($d_\perp = 2-\epsilon$):

$$
\begin{aligned}
\frac{d^3N^{\mathrm{G2\,IV}}}{d^3k} = {}& \frac{g^2}{4\pi^2}\frac{N_c}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\int_{x,y,x',y'}\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\left[-\frac{d_\perp}{\epsilon}-d_\perp\gamma_E-\frac{d_\perp}{2}\log\frac{(y'-y)^2\mu_F^2}{4}-1\right]\\
&\times\left[\int_{\frac{k^+}{\vee}}^1\frac{d\xi}{\xi^2}e^{-ik\cdot(y'-y)\frac{1}{\xi}}\left[\xi(1-\xi)+\frac{(1-\xi)}{\xi}+\xi+\frac{\xi^2}{(1-\xi)_+}\right]+\left(\log\left(1-\frac{k^+}{\vee}\right)+\frac{11}{12}\right)e^{-ik\cdot(y'-y)}\right]\\
&\times\left[U^{e'c}(y')-U^{e'c}(x')\right]\left[U^{ec}(y)-U^{ce}(x)\right]\rho^{e'}(x')\rho^{e}(x)\\
&-\frac{g^2}{4\pi^2}\frac{N_c}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\int_{x,y,x',y'}\frac{(x-y)\cdot(x'-y)}{(x-y)^2(x'-y)^2}\left[-\frac{d_\perp}{2}\log\frac{(y'-y)^2\mu_{\overline{MS}}^2}{4}\right]\left[\int_{\frac{k^+}{\vee}}^1\frac{d\xi}{\xi^2}e^{-ik\cdot(y'-y)\frac{1}{\xi}}\frac{(1-\xi)}{\xi}\right]\\
&\times\left[U^{e'c}(y)-U^{e'c}(x')\right]\left[U^{ec}(y)-U^{ce}(x)\right]\rho^{e'}(x')\rho^{e}(x)\\
&-\frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^5}\frac{1}{k^+}\left(1-\frac{k^+}{\vee}\right)\int_{x,x',z,w,w'}e^{-ik\cdot(w'-w)}\frac{(w-z)\cdot(w'-z)}{(w-z)^2(w'-z)^2}\frac{(x-z)\cdot(x'-z)}{(x-z)^2(x'-z)^2}\\
&\times\left[U^{e'c}(z)-U^{e'c}(x')\right]\left[U^{ec}(z)-U^{ce}(x)\right]\rho^{e'}(x')\rho^{e}(x)\\
&+\frac{g^2}{4\pi^2}\frac{N_c}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\frac{11}{12}\int_{x,y,x',y'}e^{-ik\cdot(y'-y)}\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\left[d_\perp\gamma_E+\frac{d_\perp}{2}\log\frac{(y'-y)^2\mu_R^2}{4}\right]\\
&\times\left[U^{e'c}(y')-U^{e'c}(x')\right]\left[U^{ec}(y)-U^{ce}(x)\right]\rho^{e'}(x')\rho^{e}(x)\\
&+\frac{g^2}{4\pi^2}\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{N_c}{k^+}\int_{\frac{k^+}{\vee}}^1 d\xi\frac{(1-\xi)}{\xi}\int_{x,y,x',y'}e^{-ik\cdot(y'-y)\frac{1}{\xi}}\left[\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}-\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2(x'-y')^2(y'-y)^2}d_\perp\right]\\
&\times\left[U^{e'c}(y')-U^{e'c}(x')\right]\left[U^{ec}(y)-U^{ce}(x)\right]\rho^{e'}(x')\rho^{e}(x).
\end{aligned}
\tag{4.1}
$$

- Lines 1–5: the DGLAP term (3.10). Lines 1–3 are your (L1)+(L4) without the pole, with $\mu\to\mu_F$ and $+\frac{11}{12}$, where $\frac{11}{12}=\frac{\beta_0}{2}\cdot\frac{1}{2N_c}$ is the $\delta(1-\xi)$ of P_{gg}. Lines 4–5 are your (L2), unchanged; its $\mu_{\overline{MS}}$ drops out for $\mathbf{k}\neq 0$.
- Lines 6–7: your (L3), unchanged.
- Lines 8–9: the running coupling (3.8), $\frac{\alpha_s\beta_0}{4\pi}=\frac{g^2N_c}{4\pi^2}\cdot\frac{11}{12}$.
- Lines 10–11: your (L5), unchanged, with $d_\perp$.

The two $\frac{11}{12}$ terms add up to $\frac{\alpha_s\beta_0}{4\pi}\frac{d_\perp}{2}\log\frac{\mu_R^2}{\mu_F^2}$LO and cancel for $\mu_R=\mu_F$. No term contains $\log\frac{\vee}{k^+}$, and the sum of lines 1–5 is finite for $\vee\to\infty$ (Step 7).